MUHAMMAD IMRAN | BT503 Environment Biotechnology

LESSON 120-140 | QUIZ NO 02

87 MCQs

1. ……techniques are best suited to instances of relatively localized pollution within a site, typically in 'hot-spots' of medium to relatively high concentration _____Ex situ
2. Which of the following method affects the reproduction of the pathogenic agents_____ UV radiation
3. A sludge with a level of dry solids of 2% has a water content of _____98%.
4. Cropping accumulater of heavy metal plants occur in ____ technology _____Extensive Technology
5. _____ is normally uses high-temperature refuse the contaminated materials_____Vetrification.
6. Which of following process does not introduce chemical product into the liquid_____Membranes.
7. Among dewatering processes the process which is least used is called_____Stabilization
8. All of these are artificial processes of pathogen removal except_____Land treatment.
9. In situ technique is suitable for_____Contramination_____Widespread.
10. A sludge with a solids load of 120 kgSS/d and a solid concentration of 2.0%n(20.000mg/L) will have a flow of_____6.0 m3/d.
11. Most frequently used unit for concentration of sludge……_____%.
12. The main characteristic of_____Methodsare that the soil is removed from where it originally lay, for treatment_____Ex situ.
13. For preliminary waste water treatment all basic equipment are required except one_____aerator.
14. The design of sludge treatment and final disposal stages is based on which of the following_____Sludge flow.
15. Which of the following pathogen removal system does not need chemicals_____Maturation ponds.
16. _____ are well suited to large scale treatment where speed is not of the essence_Extensive technologies.
17. Capillary water maintain it self absorbed in which of the following phase_____Solid.
18. In a dewatering process the dry solid level in the dewaterd sludge of filter press_____20-30%.
19. Facultative pond require how much time for sludge removal from liquid phase_____Decades.
20. The process in which decomposition of aerobic organic matter is achieved through control condition of temperature, water content, and nutrients is known as_____Compositing.
21. Following Chemicals are added in the sludge preparation process_____Coagulants and Polyelectrolytes Both a & c.
22. Soil washing and thermal treatments are example of Technologies _____Intensive Technologies.
23. Which of the following is highly recommended system for pathogen removal _____Maturation Ponds.
24. Which of the following Sludge management stage accomplished through the removal of the biodegradable organic matter of the sludge _____Stabilization.
25. A very effective method for removal of disease causing organisms is_____Ozonisation.
26. Disposal of sludge in the mining areas is known as _____Land reclamation.
27. Stages of sludge processing involve? _____ ALL
28. ----technologies have smaller resource requirement and lower initiation, running and support costs. _____ extensive technologies
29. --------- process is an appropriate method for sludge stabilization in nurseries, horticulture and landscaping._____ composting
30. _____ is very effective agent for the removal of pathogens. _____ Ozone

31. of the liquid sludge range between? _____ 1.02 and 1.03
32. If the sludge is to be incinerated then the following --------process is not necessary. _____Disinfection
33. Water in the sludge can be classified into how many classes? _____ 4
34. In primary treatment system, primary sludge takes how much time for sludge removal from liquid phase?_months
35. Lime stabilization is used to treat_____ sludge. _____ All
36. Which of following process does not introduce chemical products into the liquid._____ Membranes
37. In situ technique is suitable for ---------- contamination. _____ widespread
38. The concentration of solids in sludge is expressed in the form of? _____ Dry solids
39. ----- metallic coagulants is/are used in conditioning of sludge._____ both A&B
40. Principle mechanism of influence in the dewatering process is called -------. _____Evaporation
41. For which of the following waste water treatment system the equipment depends on the post treatment process used._____ UASB reactors+ post treatment
42. How much dry solid content is found in hard solid?_____ 35% to 60 %
43. In extensive technologies precipitationof metal sulphides takes place under------- conditions._____ anaerobic
44. The sludge production is a function of the waste water treatment system used for the_____. liquid phase
45. Time required for the accumulation of discarded sludge from WWTPs is--------___._____3 to 5 years
46. In UASB + polishing pond dry solid level ranges between?_____3-4 %
47. The pH of the medium will be ----------when lime is added to the digested sludge, in the sludge disinfection process._____ Basic
48. Among the natural dewatering processes, which is least used are called _____ sludge lagoons
49. Cropping of heavy metals accumulators plants occur in _____ Extensive methods
50. ------process is an encapsulation of contaminants within a monolithic solid of high structural integrity, with or without associated chemical fixation. _____ Vitrification
51. Which basic equipment is required for high rate pond waste water treatment process?_____ rotors of liquid movement
52. Gamma radiations are used for_____. Conditioning
53. The density of the sludge during most of its processing is very close to? _____Water
54. Density of sludge in processing stages is close to? _____ $1000\ kg/m^3$
55. Adsorbed water in the sludge can be removed by?_____ flocculating agent
56. Transformation or mineralisation of contaminants to less toxic and more mobile, forms is known as…….. remediation method. _____Biological
57. Thermal decomposition process by oxidation is known as _____. _____Incineration
58. How much water content is found in fluid sludge? _____100%-75%
59. ------- is the main water removal mechanism in the sludge dewatering…….._____ Percolation.
60. ..agent/s responsible for land contamination……._____asbestos work
61. Criteria for management of waste water treatment include……_____all
62. - The period required for carry out composition through in vessel biological reactors is normally……__14-21days
63. Normal temperature required in thermopilic aerobic digestion process is……_____50-60oC
64. ……..involves the heating of sludge at 70oC for 30 mints. _____Pasteurisation
65. Free water in the sludge can be removed by…._____gravity
66. ------is an example of thermal remediation method. _____Gasification
67. Type of centrifuges mostly used in the dewatering process are called-------. Horizontal-Shaft Centrifuge
68. Toxic by-products can be generated in which of the following system?_____ Chlorination

69. For sludge stabilization, chemical stabilization is accompanied by ------of an organic matter._____ Oxidation

70. ___is a land disposal process, in which the organic substrate is biologically degraded in the upper layer of the soil and the inorganic fraction is transformed or fixed into this layer. _____Land Farming

71. ---the following technologies have high initiation, running and support costs. Intensive technologies

72. How much dry solid content is found in fluid sludge?_____ 0%-25%

73. Thickening in sludge management is removal of --------. _____Water

74. - Water in the sludge can be divided into ------- classes. _____4

75. Sludge treatment stages are ---------. _____6

76. ---- dewatering can be done by natural or mechanical processes. _____Sludge

77. Which of following technologies key factors are a fast response and low treatment time are key factors which pertains to ---------. _____Intensive technologies

78. Virtification uses ----- temperature to fuse contaminated materials. _____High

79. In the process of dissolved air flotation release of oxygen in the form of bubbles is known as --------. _____Depressurization

80. A common way in which all form of remediation are often characterized is as -----._____ Both a & b

81. ---- is used for thickening of sludge. _____Both b& c

82. -- technologies impose less damage to soil quality. _____Extensive technologies.

83. A process in sludge management for preparation for dewatering is known as ------. _____Conditioning

84. ------- is the main natural water removal mechanism in the sludge dewatering. _____Percolation

85. Which method are costly for removal of disease causing organisms. _____Artificial

86. Removal of organic matter in sludge management is known as --------._____ Mass reduction

87. In sludge processing stages specific gravity is close to -----. _____0.1

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